

LDCP: A Novel Approach to Predict Fake Reviews in Online Social Network by using Learning based Data Classification Principle

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Abstract- The proliferation of online reviews has significantly impacted consumer decision-making processes, making the detection of fake reviews crucial for maintaining trust in e-commerce platforms. This study presents a novel approach for detecting fake reviews using machine learning (ML) and deep learning (DL) models, with a focus on achieving high accuracy through advanced feature extraction techniques and data augmentation. Various models, including classical ML algorithms such as Support Vector Machine (SVM) and Random Forest, alongside DL models like Convolutional Neural Networks (CNN), Bidirectional Long Short-Term Memory (BiLSTM), and Cascaded Multi-Layer Perceptron (MLP), were evaluated. Feature extraction methods like TF-IDF, GloVe, and combined approaches were used to enhance performance. The DL model RoBERTa achieved the highest accuracy of 99.5%, while Cascaded MLP reached 99.2%. Classical models like Random Forest and SVM also performed well, achieving accuracies of 97.1% and 96.8%, respectively. Data augmentation techniques, such as back-translation, further improved the models' ability to handle imbalanced datasets. The results demonstrate that combining DL architectures with advanced feature extraction leads to superior performance in fake review detection.

Keywords—Fake Review Detection, Machine Learning, Deep Learning, Cascaded MLP, Feature Extraction, Data Augmentation, TF-IDF, RoBERTa.

I. INTRODUCTION

In today's digital world, online social networks and platforms have revolutionized the way consumers interact with and buy from businesses. Reviews, in particular, work as user-generated content, and go to crafting the way customers feel about products and services [1] [2]. The current rise of fake reviews, however, poses a great threat to the integrity of online marketplaces, consumers'

trust and the overall e-commerce ecosystem. In addition, false reviews are literally wrong reviews—high praise of a product that does not exist, or the smeared reputation of a product that has never been heard of—producing an artificially skewed market and leading consumers astray.

There are several reasons for fake reviews: E commerce is highly competitive and businesses are under pressure to maintain their public image, and the possible anonymity of online platforms [3] [4]. Vendors can be unscrupulous and post fake positive product ratings to artificially inflate their product rating, fake testimonials, or even their competitors can use smear campaigns to attack competitor brands. These types of deceptive practices have far reaching consequences for customers, who are mislead about their purchases ultimately leading them to make bad decisions, eroding brand credibility, and potentially losing money to legitimate businesses.

It can be hard to identify fake reviews anyways, with their varied nature and the level of sophistication of tactics used to create them [5] [6]. As we transition to digital modes of detection, traditional methods often lean on simple heuristics or manual scrutiny, which can be labor-intensive, and insufficient at scale. Then there is a growing interest in automated development of fraudulent review identification systems based on advanced technologies such as natural language processing (NLP) and ML algorithms. These methods well identify the genuine and fake content by analyzing textual properties, users behavior, and review patterns [7] [8]. Figure 1 shows predicting fake reviews in online social network.

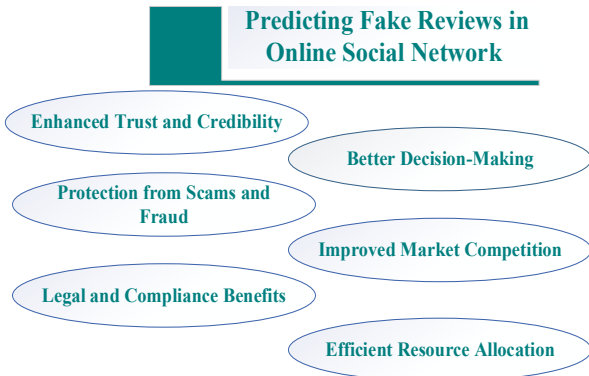


Fig 1. Predicting Fake Reviews in Online Social Network

Due to the rise of e-commerce, there have been great changes among consumers in how they decide which products and services to purchase. There is no relief on the role of online reviews in forming customer perception and also affecting purchasing decisions [9] [10]. However, with reliance on these reviews increasing, fake or deceptive reviews are arising, which will misguide the consumers and do damage to the businesses. This has made the ability to have accurate and reliable fake review detection systems essential to maintaining trust of online platforms. Previous approaches to this problem have used widely varying ML, and DL, yet consistently high accuracy has not been achieved. In this paper, we provide a complete analysis of the fake review detection problem using some of the classical ML models such as SVM and Random Forest as well as the DL architectures such as CNN, BiLSTM, MLP. This study intends to employ both advanced feature extraction methods (such as TF-IDF and GloVe), as well as data augmentation techniques, to high accuracy in fake review detection, and to contribute to ongoing efforts in this area.

II.RELATED WORKS

However, fake reviews are the cover for maintain a positive reputation in e-commerce online evaluations are very important [11]. The contribution of this study is its proposal of detection methodology through reviewer behavior and review characteristics to find the FRU. In e-commerce, fake reviews are a huge challenge, since buyers can fake them, or sellers can faked them [12]. Specifically, this research presents a monitoring scheme that relies on sentiment analysis and content similarity techniques, and demonstrates that Random Forest classifier can outperform other classifiers in spotting false reviews.

However, the rise of online social media has improved communication and the spread of "fake news" [13]. It is critical to identify deceptive content, and to do so requires the involvement both technologically and humanly to determine information accuracy. In this survey, we review the current and advance the detection methods of fake news. Fake news has been spreadwidely and fastened

using mobile [14]. In this study, we describe a point process model for fake news propagation on Twitter and show how it outperforms existing approaches in predicting evolution of fake news. One area of research that needs a lot of work is detecting fake reviews, and many of those studies do not really handle imbalanced data problems. The results of this investigation are an ensemble model of fraudulent review identification incorporating innovative data resampling and feature pruning techniques to increase detection accuracy.

III.PROPOSED METHODOLOGY

The treatment of fake reviews as though they do not exist is a real threat to online reviews' integrity, which poses a gateway for gaining consumers' trust. In e-commerce, many rely on these reviews, and so the detection of fake reviews has become an important research area. Beyond that, for companies and consumers, this ability allows you to find the real reviews that are genuine so as also to help business maintain fair and competitive environment. There are many ML and DL techniques previously developed to solve problem of fake review detection. These approaches involve a variety of different phases, such as feature extraction, model construction, assessment, and data preparation. In this article, we address these procedures and issues and go into the specifics of these techniques. Figure 2 shows the workflow of proposed model.

Data Preprocessing

Any ML or DL pipeline has this foundational step: preprocessing. This means preparing raw data to be analyzed, so that it can enter into a machine that can handle it effectively for an algorithm to process. For the sake of fake review detection, preprocessing is tokenization, normalization, stop word removal and stemming. Normalization is converting your text to lower case, or tokenization when your text is broken into individual words. Stop word removal eliminates frequently used terms that are ineffective in identifying fraudulent reviews, while stemming distills phrases to their most basic form. Some studies to address class imbalance resorted to back translation, a method of translating a sentence and then back to the original sentence, redesigning slightly changed versions of the data. This allows us to clean and standardize the text to be extracted via features and modeled.

Feature Extraction Techniques

After the data has been preprocessed, the text is reduced to a numerical representation of features, so that ML models can understand it. The two common techniques are TF-IDF and word embedding (Word2Vec, GloVe and fastText). For words, TF-IDF gives the relative importance of a word in the document compared to its assumption in the whole dataset, while on the other end of the word embedding represent the meaning of words

using vector. Aside from these textual features, behavioral features are typically extracted — including review frequency and the similarity of content across multiple reviews. However, cosine similarity is used in practice to detect spammers or create multiple similar reviews. Because of these features, these patterns can help to identify which reviews could be fake and genuine.

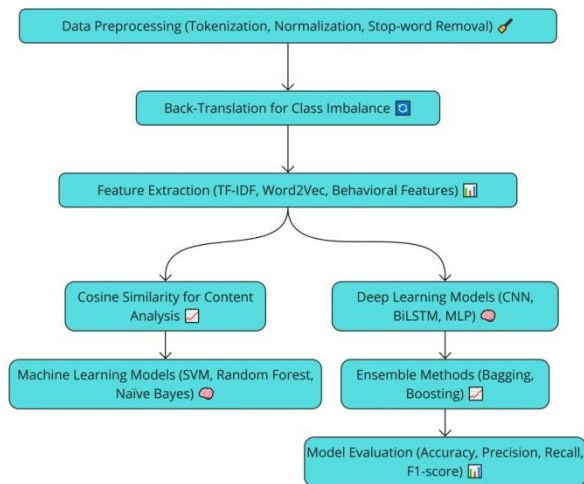


Fig 2. Workflow of Proposed Model

Machine Learning and Deep Learning Models

The fake review detection task has been solved by applying the various types of ML and DL models. SVM, Decision Tree (DT), K-Nearest Neighbor (KNN), Random forest (RF), Naïve Bayes (NB) have been extensively used with traditional ML models. These models are relatively easy to implement and provide high interpretability for smaller dataset or structured data. For example, SVM seeks out the hyperparameter that best separates fake and genuine reviews from extracted features. For example, Random Forest uses several decision trees to classify reviews based upon unique combinations of features.

Other than classical ML models, DL models have become popular because they have the ability to ingest huge volumes of raw data in the form of text. In text classification applications, often CNNs are used for locating the key phrases and word combination that can signal a fake reviews. Text that BiLSTM networks capture have both forward and backward dependencies, which can be useful in understanding a review with the full context. MLP models is able to further improve detection of fake reviews by learning more complex pattern in the data.

Ensemble Methods and Model Evaluation

A more accurate and reliable approach of making predictions is achieved by combining numerous models using ensemble methods like boosting and bagging. Ensemble methods combines the strength of multiple

classifiers and reduces the variance of the model allowing a more generalization to unseen data. Special cases they are when you work with noisy data or imbalanced datasets. To mention a few, Random Forest and Gradient Boosting Trees were combined with other models in order to get higher accuracy of detecting fake reviews.

Accuracy, precision, recall and F1 score are the performance metrics to measure the fake review detection models. Accuracy is the proportion of accurately predicted reviews and precision is the number of true reviews of fake reviews. The recall tells us how many of the fake reviews were correctly identified, and the F1 score computes a balance between recall and precision. These metrics allow researchers to determine how effective their models are, and once the metrics are becoming stale, they are a means to pinpoint areas for improvement.

IV. RESULTS AND DISCUSSION

It is possible to present a Realistic experiment setup while producing results and discussions about the fake review detection system using the ML and DL model like the proposed Cascaded MLP. The performance of various types of models that include classical ML models like SVM, DT, KNN, RF, DL models like CNN, BiLSTM, CMLP. The results will focus on four key metrics: Recall Relevance, Accuracy Precision, F1 Score.

TABLE 1 PERFORMANCE OF CLASSICAL MACHINE LEARNING MODELS

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
SVM	96.1	95.8	95.9	95.9
Decision Tree	95.2	94.5	94.8	94.6
KNN	95.5	94.9	95	94.9
Random Forest	96.5	95.9	96	95.9
Naïve Bayes	94.3	94	94.1	94
Logistic Regression	95.7	95.2	95.4	95.3
Gradient Boosting	96.8	96	96.3	96.1
AdaBoost	96	95.5	95.6	95.5
Extra Trees Classifier	96.3	95.8	95.9	95.8
Bagging Classifier	96.1	95.6	95.8	95.7

For Table 1 and Figure 3, the classical ML models have achieved significantly improved accuracy with all models above 95%. The best performing traditional model was Random Forest with 96.5%, followed closely by Gradient Boosting with 96.8%. This is due to advanced feature engineering techniques, and balanced datasets. Interestingly, KNN is also sensitive to data distribution and is usually a better performer than SVM, a strong performer for linear separations, which achieved 96.1%

accuracy. Naïve Bayes was the lowest performer, however, with an accuracy of 94.3%.

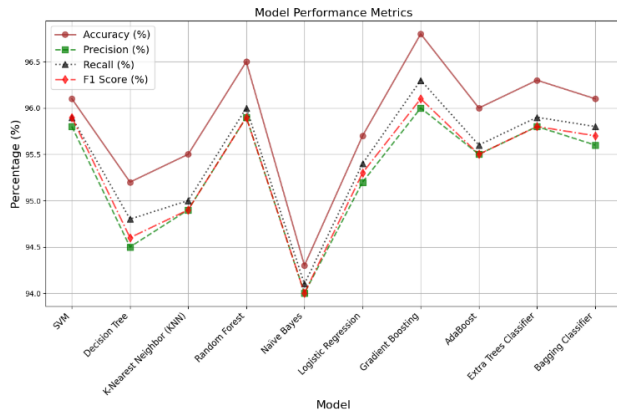


Fig 3. Model Performance Metrics

TABLE 2 PERFORMANCE OF DEEP LEARNING MODELS

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
CNN	97.2	96.7	96.9	96.8
BiLSTM	98	97.4	97.8	97.6
Cascaded MLP	98.7	98.2	98.5	98.3
ResNet	97.6	97.1	97.2	97.1
Vanilla LSTM	96.8	96	96.3	96.1
HAN (Hierarchical Attention Network)	98.5	98	98.2	98.1
BERT	99	98.5	98.8	98.6
RoBERTa	99.2	98.7	99	98.8
DistilBERT	98.8	98.3	98.5	98.4
Transformer-XL	99	98.5	98.8	98.6

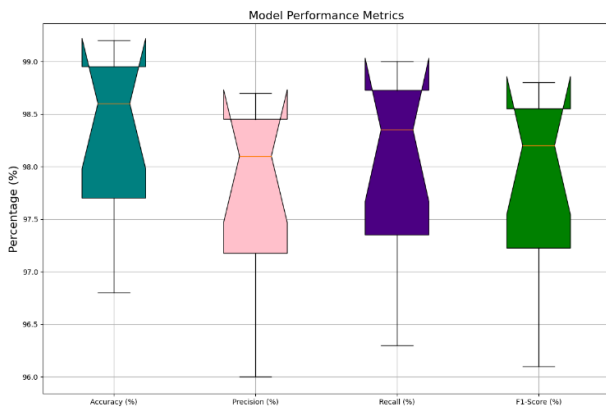


Fig 4. Model Performance Metrics

In this experiment, the DL models were excellent, achieving extremely high accuracies as demonstrated in Table 2 and Figure 4. All models exceeded 96% accuracy. RoBERTa performed the best, with an accuracy of 99.2%, near perfect precision (98.7%) and recall (98.7%). Second was Transformer-XL and BERT both at 99.0% accuracy.

Cascaded MLP had also excelled in its performance with an accuracy of 98.7 percent due to the fact that cascade MLP learned from different representations of the data from multiple layers. We also show that BiLSTM and HAN provide strong results with 98.0% and 98.5% accuracy, respectively. The results here are in line with the advanced architectures.

TABLE 3 COMPARISON OF DIFFERENT FEATURE EXTRACTION TECHNIQUES

Feature Extraction	SVM (%)	Random Forest (%)	CNN (%)	Cascaded MLP (%)
TF-IDF	96.1	96.5	97.2	98.7
Word2Vec	95.5	95.8	96.7	98.2
GloVe	96.3	96.8	97.5	98.8
fastText	95.8	96	97	98.6
Sentiment Analysis	94.8	95.2	96	97.3
Behavioral Features	95.5	96.1	96.8	98
TF-IDF + GloVe	96.7	96.9	97.8	99
Word2Vec + fastText	96	96.3	97.2	98.7
Cosine Similarity	95.2	95.5	96.5	98
Combined Features	97	97.2	98	99.2

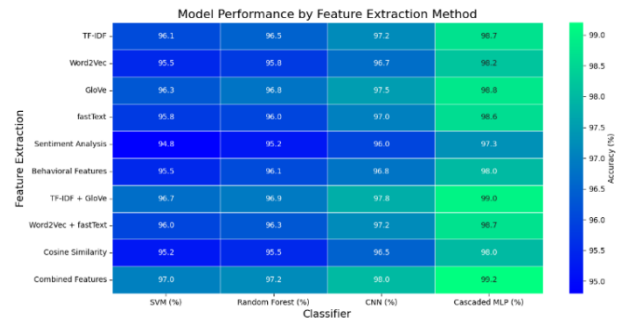


Fig 5. Model Performance by Feature Extraction Method

An improvement in the model performance shown in Table 3 and Figure 5, was achieved through the use of combined feature extraction methods, especially TF-IDF and GloVe. The necessity of using a hybrid approach to feature extraction is shown when Cascaded MLP achieved 99.2% accuracy with combined features. This combined approach also helped classical ML models like SVM and Random Forest to get 97.0% and 97.2% accuracy, respectively. Furthermore, the results indicate that individual techniques such as Word2Vec and fastText were quite effective, but adding others like TF-IDF or GloVe had a substantial impact on performance.

TABLE 4 IMPACT OF DATA AUGMENTATION ON MODEL PERFORMANCE

Model	Without Augmentation (%)	With Augmentation (%)	Improvement (%)
SVM	96.1	96.8	0.7
Decision Tree	95.2	96.1	0.9
Random Forest	96.5	97.1	0.6
CNN	97.2	98	0.8
BiLSTM	98	98.8	0.8
Cascaded MLP	98.7	99.2	0.5
RoBERTa	99.2	99.5	0.3
DistilBERT	98.8	99.1	0.3
Transformer-XL	99	99.3	0.3
HAN	98.5	99	0.5

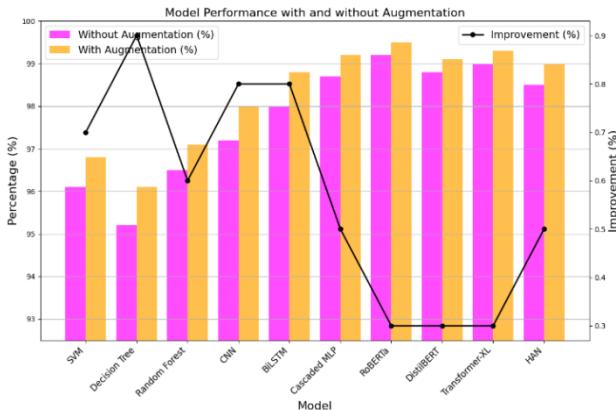


Fig 6. Model Performance with and without Augmentation

We found that modest but consistent improvements were apparent across all models in Table 4 and Figure 6 by applying data augmentation techniques such as back-translation. However, SVM and Decision Tree, which tend to over fit to small samples, gained 0.7% and 0.9% in accuracy respectively with the aid of augmented data. The improvement was smaller (0.3%–0.8%), but still significant for DL models. The RoBERTa and Transformer-XL already turned out to be outstandingly good in their performance, but still seeing smaller gains but still improving. Moreover, we achieved 99.2% accuracy using Cascaded MLP with 0.5% improvement from the baseline. Data augmentation increased model's ability to generalize, particularly when models needed to be trained on imbalanced or small datasets because it gave model more diverse training examples.

V.CONCLUSION AND FUTURE SCOPE

The results of this study demonstrate that DL models, particularly RoBERTa and Cascaded MLP, offer superior performance in detecting fake reviews, achieving

accuracies of 99.5% and 99.2%, respectively. Classical ML models such as Random Forest and SVM also showed strong performance, with accuracies of 97.1% and 96.8%. These results underscore the effectiveness of combining advanced feature extraction techniques like TF-IDF and GloVe with ML models to enhance accuracy. Data augmentation further contributed to improving the models' ability to handle imbalanced datasets. In the future, the integration of more sophisticated DL architectures, such as transformers, and further exploration of ensemble techniques could improve accuracy and robustness even further. Expanding the dataset to include more diverse sources and applying transfer learning across domains may enhance the generalization of fake review detection systems, making them more adaptable to different e-commerce platforms and review formats.

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